

module2d3

☒ VLANs & Inter-VLAN Routing

1. ☆ What is a VLAN?

VLAN (Virtual Local Area Network) logically divides one physical switch network into multiple separate **broadcast domains**.

Why use VLANs?

- 🛡️ Improve security
- 📁 Reduce broadcast traffic
- □ Separate departments/users
- ⚙️ Make network management easier
- □ Improve network performance

Important: Devices in different VLANs **cannot communicate directly at Layer 2**. They need **Layer 3 routing**.



2. ☆ VLAN Types

VLAN	Purpose	Key Point
Default VLAN	Initial switch VLAN	VLAN 1; cannot be deleted/renamed
Data VLAN	User data	Web, email, etc.

VLAN	Purpose	Key Point
Management VLAN	Management traffic	SSH, Telnet, SNMP
Native VLAN	Untagged trunk traffic	Default is VLAN 1
Voice VLAN	VoIP	Supports QoS
Black Hole VLAN	Unused ports	Security purpose

☆ Remember:

Native VLAN = untagged traffic

Management VLAN = manage the switch

Voice VLAN = IP phones

Data VLAN = normal user traffic



3. ☆ Access Port vs Trunk Port

Access Port

Carries traffic for **ONE VLAN**.

Usually connected to:

- PC
- Printer

- Server

```
switchport mode access  
switchport access vlan 10
```

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Trunk Port

Carries traffic for **MULTIPLE VLANs**.

Usually connects:

- Switch ↔ Switch
- Switch ↔ Router
- Switch ↔ Multilayer switch

```
switchport mode trunk
```

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4. ☆ IEEE 802.1Q

802.1Q is the standard used to identify VLAN traffic on a trunk.

It adds a **4-byte VLAN tag** to an Ethernet frame.

Remember:

Native VLAN

Traffic belonging to the **native VLAN** is normally sent **untagged** across an 802.1Q trunk.

By default:

Native VLAN = VLAN 1

⚠ The native VLAN should match on both ends of a trunk.



5. ☆ DTP

DTP = Dynamic Trunking Protocol

Cisco proprietary protocol used to **negotiate trunking** between switches.

For security, user-facing ports should normally be configured explicitly as access ports and DTP disabled where appropriate:

```
switchport mode access  
switchport nonegotiate
```

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6. ☆ VLAN Ranges

Range	Name	Important
1–1005	Normal	VLAN database (vlan.dat)
1006–4094	Extended	Stored in running configuration

⚠ 1002–1005 are reserved VLANs.



7. ☆ Important VLAN Commands

Create VLAN

```
vlan 10  
name SALES
```

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Assign port to VLAN

```
interface fa0/1  
switchport mode access  
switchport access vlan 10
```

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Configure trunk

```
interface fa0/24  
switchport mode trunk
```

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Configure native VLAN

```
switchport trunk native vlan 99
```

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Allow specific VLANs

```
switchport trunk allowed vlan 10,20,30
```

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Add VLAN without replacing existing allowed VLANs

```
switchport trunk allowed vlan add 300
```

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8. ☆ Verification Commands

View VLANs

```
show vlan brief
```

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View trunk information

```
show interfaces trunk
```

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View configuration

```
show running-config
```

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View interface IP/status

```
show ip interface brief
```

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9. ☆ VLAN Hopping

VLAN hopping = an attack that allows an attacker to access traffic belonging to another VLAN.

Two important methods:

A. Switch Spoofing

Attacker sends **DTP messages** and tricks the switch into creating a trunk.

Protection:

```
switchport mode access  
switchport nonegotiate
```

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B. Double-Tagging

Attacker sends a frame containing **two VLAN tags**.

The attack takes advantage of the **native VLAN** behavior.

Protection:

- Don't use VLAN 1 as native VLAN
- Use an unused VLAN as native VLAN
- Keep native VLAN consistent
- Properly configure trunks



10. ☆ Inter-VLAN Routing

Inter-VLAN routing allows devices in **different VLANs** to communicate.

Example:

```
PC-A
VLAN 10
|
Switch
|
Router/L3 Switch
|
Switch
|
PC-B
VLAN 20
```

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Without routing:

× VLAN 10 → VLAN 20

With routing:

□ VLAN 10 → Router/L3 Switch → VLAN 20



11. ☆ Three Inter-VLAN Routing Methods

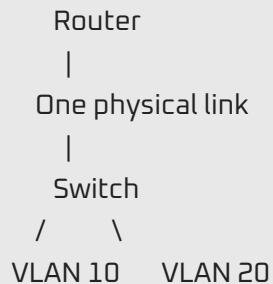
Method	Main Idea	Important
Legacy Routing	One physical router interface per VLAN	Poor scalability
Router-on-a-Stick	One router interface + subinterfaces	Uses 802.1Q
Multilayer Switching	Layer 3 switch + SVIs	Fast and scalable



12. ☆ Router-on-a-Stick (ROAS)

ROAS uses **one physical router interface** divided into multiple **subinterfaces**.

Example:



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☆ Key word:

ROAS → Subinterfaces → 802.1Q

13. ☆ SVI

SVI = Switch Virtual Interface

Created using:

```
interface vlan 10
```

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An SVI provides a **Layer 3 interface for a VLAN** and can serve as the **default gateway** for hosts in that VLAN.

Example:

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
```

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Hosts in VLAN 10 can use:

Default Gateway = 192.168.10.1

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14. ☆ Multilayer Switch

A multilayer switch can perform **Layer 2 switching AND Layer 3 routing**.

Enable routing:

```
ip routing
```

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Then create SVIs:

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
```

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Now the switch can route:

```
VLAN 10 ↔ VLAN 20
```

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☆ Very important:

SVI + ip routing = Inter-VLAN routing on a multilayer switch



15. ☆ SVI vs Routed Port

SVI

```
interface vlan 10
```

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- Virtual interface
- Represents a VLAN
- Can act as default gateway

Routed Port

```
interface g0/1  
no switchport  
ip address 10.0.0.1 255.255.255.252
```

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- Physical Layer 3 interface
- Does not operate as a normal Layer 2 switchport
- Commonly used for Layer 3 links between network devices

Remember:

SVI = Virtual

Routed port = Physical Layer 3 interface



16. ☆ CEF

CEF = Cisco Express Forwarding

It is Cisco's high-performance packet-forwarding mechanism.

Main idea:

CEF → Fast Layer 3 packet forwarding



17. ☆ Implementing a New VLAN – 5 Steps

Memorize this sequence:

1 Create VLAN

```
vlan 300  
name NEW_VLAN
```

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2 Identify ports

Determine:

- Access ports

- Trunk ports

3 Configure trunks

Make sure VLAN 300 is allowed:

```
switchport trunk allowed vlan add 300
```

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4 Create SVI

```
interface vlan 300  
ip address 192.168.30.1 255.255.255.0  
no shutdown
```

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5 Verify

```
show vlan brief  
show interfaces trunk  
ping
```

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18. ☆ Troubleshooting Inter-VLAN Routing

When VLAN communication doesn't work, check these **4 things**:

① VLAN

Check:

```
show vlan brief
```

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Ask:

- Does VLAN exist?
- Is the port assigned correctly?
- Does the VLAN exist on the necessary switches?

② SVI

Check:

- Correct IP?
- Correct subnet mask?
- Correct VLAN number?
- Is SVI **up/up**?

③ Routing

On multilayer switch:

```
show running-config
```

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Make sure:

```
ip routing
```

 Copy

is enabled.

④ Host

Check:

- IP address
- Subnet mask
- Default gateway

